Geographic Information Technologies and Remote Sensing in Exploratory Spatial Data Analysis: Review and Empirical Application to Smallholding in Peru

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Abstract. This literature review systematically examines the main Geographic Information Technologies (GIT) and Remote Sensing techniques, analyzing their fundamental characteristics, practical utility, areas of applicability, and degree of complementarity in the context of exploratory spatial data analysis (ESDA). The review is structured around a systematic search of academic databases and specialized literature, covering key concepts in descriptive spatial statistics —including the mathematical formulation of Moran and Geary spatial autocorrelation indices—, geographic data representation paradigms (vector and raster), coordinate reference systems, relational geodatabase management, spatial data preprocessing, and spatial pattern analysis methods. Open-source software tools such as QGIS, GeoDa, PostgreSQL/PostGIS, GDAL/OGR, and OpenRefine are examined. As an illustrative application, a case study is presented on the spatial autocorrelation of parcel fragmentation at the departmental level in Peru, using microdata from the National Agricultural Survey (ENA 2014–2024). The results reveal significant positive spatial autocorrelation ($I = 0.253$, $p = 0.018$), evidencing clusters of high fragmentation (smallholding) in the central and southern highlands, against clusters of greater parcel consolidation on the coast and in the Amazon. It is concluded that the synergistic integration of these technologies constitutes a robust and indispensable methodological approach for understanding and modeling phenomena with spatial dependence.

Keywords: Spatial statistics · Geographic information systems · Remote sensing · Spatial autocorrelation · Moran's index · GeoDa · QGIS · PostGIS · Exploratory spatial data analysis

1 Introduction

The emergence and consolidation of Geographic Information Technologies (GIT) over the past five decades has profoundly transformed the way science approaches the study of phenomena distributed in geographic space. From the first Geographic Information Systems (GIS) developed in the 1960s —particularly the

Canada Geographic Information System (CGIS), considered the first operational GIS [40]— to current cloud-based spatial computing environments, technological evolution has exponentially expanded the available analytical capabilities [30].

Spatial statistics, understood as the set of quantitative methods that explicitly incorporate the geographic dimension of data, emerges as an interdisciplinary field that articulates geography, mathematics, computer science, and Earth sciences [1,14]. Its contemporary relevance spans from spatial epidemiology [26] to urban planning and environmental monitoring [17].

In parallel, Remote Sensing has experienced unprecedented development with the proliferation of satellite platforms and the massive availability of freely accessible images —such as the Landsat, Sentinel, and MODIS series [16,43]—.

In this context, this review critically examines the characteristics, utility, applicability, and complementarity of the different GITs and remote sensing techniques, articulating the methodological foundations of ESDA. The specific objectives of the review are:

1. Systematize the fundamental concepts of descriptive spatial statistics and their mathematical formulations.
2. Characterize the types of spatial data (vector and raster) and their methodological implications.
3. Examine coordinate reference systems, datums, and cartographic projections.
4. Analyze tools for spatial database management and preprocessing of geographic information.
5. Review spatial pattern analysis methods and cartographic visualization techniques.
6. Evaluate the complementarity between GITs and remote sensing techniques.

1.1 Novelty and Contribution

Most of the literature reviewed here treats geographic information technologies and spatial statistics as if they belonged to separate disciplines: GIS textbooks rarely go beyond a passing mention of Moran's I [31], and spatial-econometrics texts [14] seldom discuss vector/raster data models or coordinate reference systems in any depth. This review brings both sides together into a single pipeline, organized around six thematic axes that follow the order an analyst actually works through on a real project, from raw data to geodatabase management, preprocessing, pattern analysis, and remote sensing integration. A second distinguishing feature is that every tool discussed is open source (QGIS, GeoDa, PostgreSQL/PostGIS, GDAL/OGR, PySAL, OpenRefine), which stands in contrast to a GIS literature still largely centered on proprietary platforms such as ArcGIS.

The clearest gap, however, only became apparent once the search moved from general spatial-statistics methodology toward applications specific to Peru: no published study had applied global Moran's I or LISA statistics to the National Agricultural Survey (ENA 2014–2024) at the departmental level. This review closes that gap with a full-scale empirical validation over the complete pooled dataset (14.7 GB, 3.8 million records), confirming through formal significance testing that smallholding in the Andean highlands is not a scattered phenomenon but a spatially clustered one. Together, the technology-integration framework and this empirical validation give the review a concrete, reproducible template that future work on provincial- or district-level ESDA in Peru can build on directly.

2 Review Methodology

The review was conducted following the guidelines for systematic narrative reviews proposed by Green et al. [22]. The search strategy was implemented in Scopus, Web of Science, Google Scholar, and institutional repositories, using key terms such as *"spatial statistics"*, *"GIS"*, *"remote sensing"*, *"ESDA"*, *"spatial autocorrelation"*, *"geodatabase"*, and *"spatial data quality"*.

Inclusion criteria considered: (a) publications in indexed journals, reference books, and international standards; (b) direct relevance to the theoretical and methodological foundations of ESDA; (c) publications in English or Spanish; and (d) priority given to recent editions of classic works and seminal contributions to the field. Articles of pure application without significant methodological contribution were excluded. In total, more than 50 bibliographic sources were reviewed, organized into six thematic axes.

3 Foundations of Descriptive Spatial Statistics

3.1 Definition and Conceptual Scope

Spatial statistics is distinguished by incorporating the location and neighborhood relationships between observations as explicit components of the analysis. Unlike classical statistics (i.i.d. assumption), it recognizes that geographic phenomena exhibit **spatial autocorrelation**: the systematic similarity between values at nearby locations.

This phenomenon is synthesized in the First Law of Geography, formulated by Tobler [39]: *"everything is related to everything else, but near things are more related than distant things"*. Cressie [14] distinguishes three major paradigms: (1) geostatistical data (continuous variables in continuous space), (2) areal or lattice data (observations on discrete units), and (3) point patterns (locations of events in space).

3.2 Main Indicators and Mathematical Formulations

Geographic Mean Center. The centroid of n points is calculated as:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} y_i \qquad (1)$$

where (x_i, y_i) are the coordinates of point i. The **spatial median center** — which minimizes the sum of Euclidean distances— is a more robust alternative [42].

Standard Distance. Quantifies the dispersion of the point set relative to the mean center:

$$SD = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{X})^2 + \sum_{i=1}^{n}(y_i - \bar{Y})^2}{n}} \tag{2}$$

The **Standard Deviational Ellipse** (SDE) generalizes this measure by incorporating the predominant orientation of the spatial pattern [27].

Global Moran's I Index. Moran's I statistic [31] is the most widely used measure of global spatial autocorrelation:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(z_i - \bar{z})(z_j - \bar{z})}{\sum_i (z_i - \bar{z})^2} \tag{3}$$

where $\bar{z}$ is the mean of the variable and w_{ij} the spatial weight between locations i and j. The expected value under the null hypothesis is $E[I] = -1/(n-1)$ [11].

Geary's C Index. An alternative measure based on squared differences [19]:

$$C = \frac{(n-1)}{2\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(z_i - z_j)^2}{\sum_i (z_i - \bar{z})^2} \tag{4}$$

Its expected value under randomness is $E[C] = 1$; values $C < 1$ indicate positive autocorrelation.

LISA Statistics. *Local Indicators of Spatial Association* (LISA) [2] decompose the global statistic into local contributions:

$$I_i = \frac{(z_i - \bar{z})}{\sum_k (z_k - \bar{z})^2 / n} \sum_j w_{ij}(z_j - \bar{z}) \tag{5}$$

LISA values are represented in **LISA maps**, classifying each unit as: *High-High* (HH), *Low-Low* (LL), *High-Low* (HL), and *Low-High* (LH).

Kernel Density Estimation (KDE). A nonparametric technique that estimates the intensity $\hat{\lambda}(s)$ of a point pattern [7,35]:

$$\hat{\lambda}(s) = \sum_{i=1}^{n} \frac{1}{h^2} K\left(\frac{s - s_i}{h}\right) \tag{6}$$

where $K(\cdot)$ is the kernel function and h the bandwidth parameter.

3.3 Spatial Weights Matrices

Spatial weights matrices **W** formalize the neighborhood structure between observations. Construction criteria include contiguity (*rook*, *queen*), distance (k-NN, fixed radius, inverse distance weighting), and row standardization [1,20].

3.4 Software for Exploratory Spatial Analysis

GeoDa [5] integrates descriptive spatial analysis, interactive visualizations with *brushing and linking*, calculation of autocorrelation indices, and spatial econometrics models. R —with packages `spdep` [8], `sf`, `spatstat`— and Python —with PySAL [34], GeoPandas, Rasterio— provide highly flexible programming environments.

4 Types of Spatial Data: Vector and Raster

4.1 The Vector Data Model

The vector model represents geographic space through points, lines, and polygons according to the ISO 19125 standard (*Simple Features*) [33,30]. The most widely used formats include Shapefile (ESRI), GeoPackage (OGC), GeoJSON, and GML/KML.

4.2 The Raster Data Model

The raster model divides space into a regular grid of cells, with **spatial resolution** determining the level of detail [10]. Relevant formats: GeoTIFF, NetCDF, Cloud Optimized GeoTIFF (COG), and HDF5.

4.3 Comparison and Complementarity

Table 1 summarizes the main differences between both paradigms.

Table 1. Comparison between spatial data models

Criterion	Vector	Raster
Representation	Points, lines, polygons	Cell grid
Phenomena	Discrete, bounded	Continuous, gradients
Geometric prec.	High (exact coordinates)	Limited by resolution
Attributes	Multiple and complex	Generally one per band
Operations	Overlay, buffer, intersection	Map algebra
Formats	Shapefile, GeoPackage, GeoJSON	GeoTIFF, NetCDF, COG

Source: Own elaboration based on [30,10].

5 Coordinate Reference Systems, Datums, and Projections

5.1 Geographic and Projected Systems

A Coordinate Reference System (CRS) defines the mathematical framework for assigning unique coordinates to the Earth's surface [37]. They are classified into: **Geographic Coordinate Systems (GCS)** —express position in latitude and longitude (e.g., EPSG:4326)— and **Projected Coordinate Systems (PCS)** —transform the surface into a 2D plane with metric units (e.g., UTM Zone 18S, EPSG:32718)—.

5.2 Geodetic Datum

Modern geocentric datums —WGS84, GRS80, SIRGAS2000— are adjusted to the Earth's center of mass and are compatible with GNSS/GPS. The PROJ library automatically manages transformations between datums (Helmert, Molodensky, NTv2 grids).

5.3 UTM Projection and EPSG Registry

The UTM projection divides the surface into 60 zones of 6° with scale distortion below 0.04% at the central meridian ($k_0 = 0.9996$) [37]. The EPSG registry provides standardized codes for identifying CRS, datums, and transformations.

6 Spatial Data Quality

The ISO 19157 standard [25] defines quality dimensions: completeness, logical consistency, positional accuracy, thematic accuracy, temporal quality, and usability. Among the most frequent errors in vector data are **topological errors**: overlapping or gapped polygons, lines with undershoot/overshoot, and invalid geometries. QGIS and PostGIS provide validation functions such as `ST_IsValid()`, `ST_MakeValid()`, and `ST_IsValidReason()`.

7 Spatial Database Management

7.1 PostgreSQL/PostGIS

PostGIS extends PostgreSQL by adding support for storing, indexing, and querying geographic data [32]. It organizes measurement functions (`ST_Distance`, `ST_Area`), spatial relationship functions (`ST_Intersects`, `ST_Contains`), geoprocessing functions (`ST_Buffer`, `ST_Union`), and CRS management (`ST_Transform`). Indexing using R-trees (GiST) reduces filtering complexity from $O(n)$ to $O(\log n)$.

7.2 Geographic ETL Processes

GDAL/OGR constitutes the reference library for conversion between more than 200 geospatial formats, being used by QGIS, GRASS GIS, and multiple platforms. The `ogr2ogr` and `gdalwarp` utilities are fundamental tools in the ETL workflow.

8 Spatial Data Preprocessing

Preprocessing includes: spatial k-NN imputation, imputation by mean/median of adjacent units, geostatistical methods based on kriging [14], and geographically weighted regression (GWR) [18]. Hierarchical Bayesian conditional autoregressive (CAR) models offer a further alternative for jointly modeling spatial dependence and covariate effects, as recently demonstrated for agricultural yields and climate vulnerability in southern Peru [29]. **Spatial outliers** require detection through LISA statistics and Moran scatter plots [2].

OpenRefine is useful for toponym normalization (clustering algorithms based on Levenshtein distance), reconciliation with external databases (Wikidata, GeoNames), and programmatic transformations using GREL.

9 Spatial Pattern Analysis and Cartographic Visualization

9.1 Choropleth Maps and Classification Methods

Choropleth maps represent quantitative data through color gradation. The most commonly used classification methods are: equal intervals, quantiles, Jenks natural breaks, and standard deviation [36].

9.2 GeoDa and Visualization in QGIS

GeoDa integrates LISA maps with significance by permutations, spatial correlograms, and Moran scatter plots with brushing and linking [5,3]. QGIS provides advanced thematic symbology and the **Print Layout** for composing maps with standard cartographic elements [9].

Table 2 summarizes the tools reviewed.

10 Remote Sensing: Characteristics and Complementarity with GITs

10.1 Foundations

Remote sensing obtains information about objects through reflected or emitted electromagnetic radiation [28]. Systems are characterized by four resolutions: spatial (pixel size), temporal (revisit frequency), spectral (number of bands), and radiometric (sensor sensitivity). They are classified as **passive** (Landsat, Sentinel-2, MODIS) and **active** (SAR, LiDAR).

Table 2. Software tools for spatial statistical analysis

Software	**Type**	**Functionality**	**Application**
QGIS	Desktop GIS	Management and visualization	Cartography, GIS
GeoDa	ESDA	Exploratory analysis	Moran, LISA
PostGIS	Spatial DB	Spatial SQL queries	ETL, queries
R/Python	Programming	Programmatic analysis	Modeling
OpenRefine	Cleaning	Tabular normalization	Preprocessing
GDAL/OGR	Library	Format conversion	ETL, CRS

Source: Own elaboration.

10.2 Spectral Indices

The **Normalized Difference Vegetation Index** (NDVI) [41] is the most widely used:

$$NDVI = \frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + \rho_{Red}} \tag{7}$$

ranging between -1 and $+1$, with values > 0.6 indicative of dense vegetation.

10.3 Main Satellite Missions

Table 3 summarizes freely accessible missions.

Table 3. Main terrestrial observation satellite missions

Mission	**Agency**	**Sp. res.**	**Bands**	**Revisit**
Landsat 8/9	USGS/NASA	30 m (15 m pan)	11	16 days
Sentinel-2A/B	ESA	10–60 m	13	5 days
MODIS	NASA	250 m–1 km	36	1–2 days
SRTM	NASA/NGA	30 m (DEM)	–	Single
Sentinel-1A/B	ESA	5×20 m SAR	C	6 days

Source: Own elaboration based on [16,44].

10.4 Integration of Remote Sensing and GIS

The standard processing chain comprises: (1) radiometric correction, (2) atmospheric correction (DOS, FLAASH, Sen2Cor), (3) geometric correction and orthorectification, (4) supervised classification (Random Forest, SVM) or unsupervised (K-Means), (5) validation using confusion matrices and Cohen's kappa statistic [12], and (6) integration into spatial analysis systems [28].

11 Case Study: Spatial Autocorrelation of Farm Parcel Structure in Peru (ENA 2014–2024)

11.1 Context and Data

The National Agricultural Survey (ENA), conducted by INEI and MIDAGRI [15], constitutes the main instrument for monitoring the country's agricultural dynamics. Because ENA observations exhibit pronounced spatial autocorrelation, standard random train/test splits can yield overly optimistic validation metrics; blocked and hierarchical cross-validation schemes have recently been proposed specifically for this dataset to mitigate such bias [6]. The case study uses: (a) vector layer of departmental boundaries in GeoPackage (EPSG:4326) [23]; (b) processed microdata from `ENA_2014_2024.sav` (14.7 GB), extracting variable P103 (number of parcels) weighted by expansion factor; and (c) DEM derived from SRTM (30 m) processed in Google Earth Engine [21].

11.2 Applied Methodology

Aggregation of the SPSS database was performed with Python (`pyreadstat`, `pandas`). A recurring practical obstacle at this stage was that department names in the `.sav` file did not always match the boundary layer's names exactly: accented characters (ñ, í, ó) were encoded inconsistently between the two sources, which silently broke the attribute join for several departments until each mismatch was traced and corrected by hand. Only after this cleanup step did the join reach a full 100% match across all 25 departments. The vector layer was reprojected to UTM Zone 18S (EPSG:32718) and topologically verified with the QGIS *Geometry Checker*. Autocorrelation analysis was conducted with Python (`libpysal`, `esda`) and cross-validated in GeoDa [5]: a first-order *queen* contiguity weights matrix (row-standardized) was constructed from the 25 Peruvian departmental boundaries, and the **global Moran's I** (999 permutations) and **LISA statistics** were calculated. Figure 1 summarizes the complete processing pipeline.

11.3 Results

Descriptive Statistics. At the national level, the departmental average number of parcels per agricultural unit shows a mean $\bar{x} = 0.236$ and standard deviation $s = 0.110$ (full pooled dataset, $n = 3{,}843{,}598$ records, 25 departments). Table 4 presents the five departments with the highest raw fragmentation ratio, and Table 5 reports the departments identified as statistically significant HH/LL clusters by the local LISA analysis, together with their individual permutation-based p-values.

Global Spatial Autocorrelation. Moran's I yielded $I = 0.253$ (expected value $E[I] = -0.042$), with $p = 0.018$ (999 permutations, $z = 2.315$). H_0 is rejected at $\alpha = 0.05$, confirming **significant positive spatial autocorrelation** across Peruvian departments.

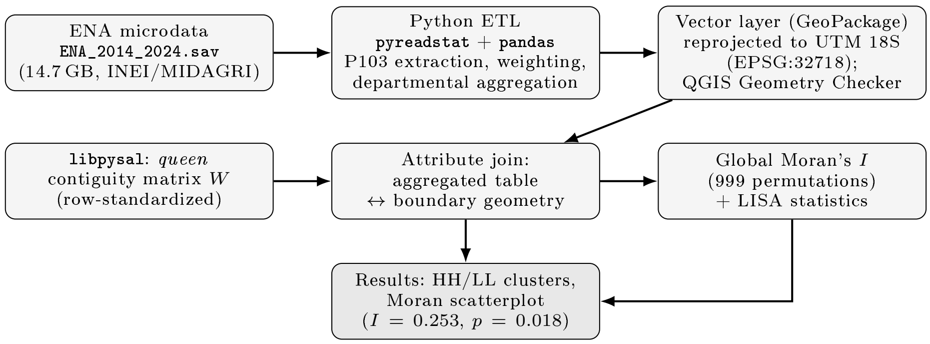

Fig. 1. Methodological workflow of the case study, from raw ENA microdata to the spatial autocorrelation results.

Table 4. Top 5 departments with highest fragmentation (smallholding)

Department	Total Parcels	Total Units	Average
Moquegua	49,728	122,830	0.405
Puno	7,695,944	19,625,840	0.392
Junin	3,034,302	7,851,223	0.386
Huancavelica	2,424,650	6,481,236	0.374
Pasco	811,001	2,280,654	0.356

Table 5. Statistically significant LISA clusters and their local permutation p-values

Cluster type	Department	Local p-value
High-High (HH)	Junin	0.009
	Huancavelica	0.013
	Ayacucho	0.023
	Apurimac	0.029
Low-Low (LL)	Piura	0.020
	Cajamarca	0.027
	Amazonas	0.035

Source: Own processing with `libpysal`/`esda` (999 conditional permutations per unit).

LISA Analysis. The local analysis ($p < 0.05$) revealed: *High-High (HH) Cluster* in Junin ($p = 0.009$), Huancavelica ($p = 0.013$), Ayacucho ($p = 0.023$), and Apurimac ($p = 0.029$) in the central and southern highlands; *Low-Low (LL) Cluster* in Piura ($p = 0.020$), Cajamarca ($p = 0.027$), and Amazonas ($p = 0.035$) on the northern coast and Amazon region. Table 5 lists these significant clusters with their local permutation p-values, and Figure 2 presents the Moran scatterplot derived from the complete ENA 2014–2024 dataset (3.8 million records). Table 6 summarizes the global findings.

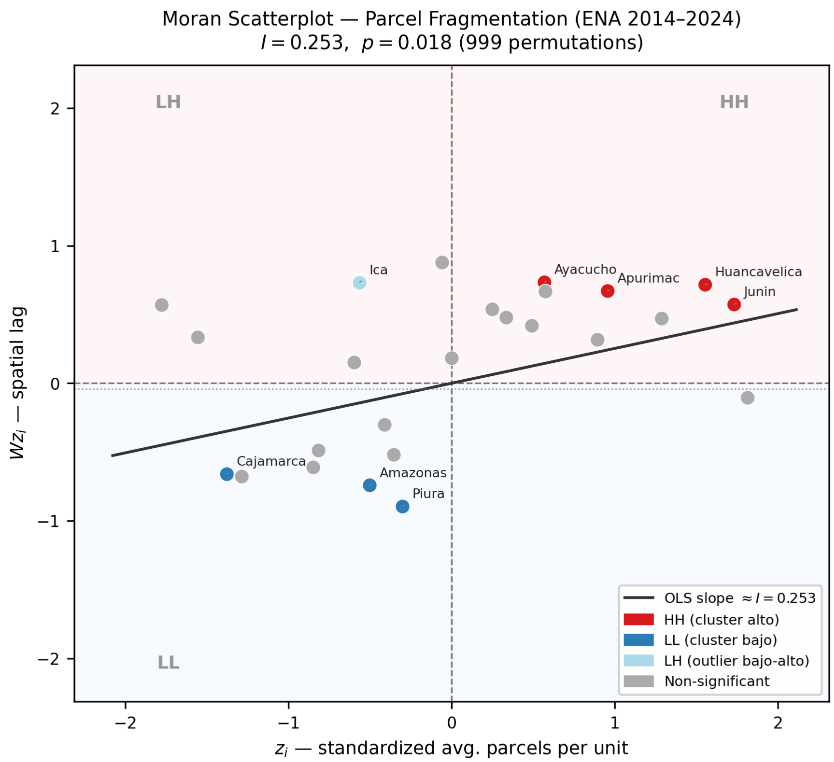

Fig. 2. Moran scatterplot of parcel fragmentation across Peruvian departments (ENA 2014–2024, $n = 25$). Red points: High-High clusters (HH); blue points: Low-Low clusters (LL); grey: non-significant. Slope of the OLS regression line equals Moran's $I = 0.253$ ($p = 0.018$, 999 permutations). Generated with `libpysal` and `esda`.

11.4 Case Study Discussion

Processing 3.8 million records (14.7 GB) from the pooled ENA 2014–2024 dataset with Python confirms the robustness of combining GITs and Data Science. The

Table 6. Empirical results of parcel spatial autocorrelation

Indicator	**Value**
Mean avg. parcels ($\bar{x}$)	0.236
Standard deviation (s)	0.110
Highest fragmentation dept.	Moquegua (0.405)
Lowest-fragmentation LL cluster	Amazonas ($p = 0.035$)
Global Moran's I	0.253 ($p = 0.018$, $z = 2.315$)
HH Clusters ($p < 0.05$)	Junin, Huancavelica, Ayacucho, Apurimac
LL Clusters ($p < 0.05$)	Amazonas, Cajamarca, Piura

Source: Own processing with Python and GeoDa, ENA data [15].

global Moran's I of 0.253 ($p = 0.018$) confirms statistically significant positive spatial autocorrelation: departments with high parcel fragmentation tend to be surrounded by similarly fragmented neighbors. The smallholding phenomenon (HH) is not geographically isolated but anchored in the south-central Andean terrain (Junin–Apurimac–Huancavelica–Ayacucho axis). Conversely, LL clusters in Amazonas, Cajamarca, and Piura reflect greater parcel consolidation on the northern fringe. Limitations include the large intra-departmental variance that departmental aggregation fails to capture, and the pooling across survey years, which may attenuate annual autocorrelation signals. Scaling to provincial or district level in future work is strongly recommended.

12 General Discussion and Perspectives

12.1 Complementarity and Integration of GITs

The different GITs constitute complementary components of an integrated methodological ecosystem that articulates: (1) storage in geodatabases (PostgreSQL/PostGIS), (2) data cleaning (OpenRefine, GDAL/OGR), (3) visual exploration of patterns (GeoDa), (4) advanced modeling (QGIS, R/Python), and (5) remote sensing integration [17].

12.2 Emerging Trends

Contemporary geospatial analysis is undergoing transformations driven by: cloud spatial computing (Google Earth Engine [21], AWS Open Data), geospatial machine learning (CNN, semantic segmentation), probabilistic latent-state models such as hidden Markov models for uncovering spatial latent heterogeneity in agricultural sustainability [13], GeoAI (*Geospatial Artificial Intelligence*), democratization of open data (OpenStreetMap, Copernicus), and territorial digital twins. However, this democratization poses significant challenges in the critical evaluation of source quality.

12.3 Review Limitations

The main limitations are: (a) as a narrative review, no formal PRISMA protocol was applied; (b) focus was on open-source tools, limiting the analysis of proprietary solutions (ArcGIS Pro, ENVI); (c) the rapid evolution in GeoAI may render some references outdated; and (d) ethical and privacy aspects of geospatial data were not addressed.

13 Conclusions

This review has examined the ecosystem of GITs and Remote Sensing techniques for ESDA, complemented with an applied case study on the agricultural parcel structure in Peru. The main conclusions are:

1. Spatial statistics —grounded in Moran's I and Geary's C statistics— provides the methodological framework for analyzing phenomena with spatial dependence.
2. Vector and raster models are complementary paradigms whose selection depends on the nature of the modeled phenomenon.
3. CRS, datums, and cartographic projections constitute the indispensable mathematical scaffolding for multi-source data integration.
4. Management in geodatabases (PostgreSQL/PostGIS) with rigorous ETL and ISO 19157 controls provides the basis for reproducible and scalable analyses.
5. Open-source tools (QGIS, GeoDa, PostgreSQL/PostGIS, GDAL/OGR, R/Python) form a mature and interoperable ecosystem for the complete geospatial workflow.
6. Remote sensing exponentially expands observational capabilities; integration with ESDA is indispensable in modern territorial analysis.
7. The ENA-Peru case study empirically confirms the deep territorial roots of Andean smallholding (Junin–Apurimac–Huancavelica–Ayacucho axis; $I = 0.253$, $p = 0.018$), demonstrating the operability of a fully reproducible Python workflow (`libpysal`, `esda`) applied to 3.8 million records across 14.7 GB of ENA microdata.

Future lines of work include: extension of the analysis to municipal scale, multitemporal ENA 2014–2024 analysis, comparative evaluation of supervised classifiers on Sentinel-2, exploration of Google Earth Engine, integration of SAR/LiDAR data with ESDA, and adoption of more advanced spatial-statistical frameworks recently applied to Peruvian agricultural data, including hierarchical Bayesian CAR models [29], spatially-blocked cross-validation schemes [6], and hidden Markov models for latent spatial heterogeneity [13].

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